SDEV 2401

Rapid Backend App Development

REST API Development - 1

A LEADING POLYTECHNIC COMMITTED TO YOUR SUCCESS

Expectations - What I expect from you

- No Late Assignments

- No Cheating

- Be a good classmate

- Don’t waste your time

- Show up to class

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Agenda

On the right is what we will cover today.

- ModelSerializers in DRF
- Routers and URL Patterns
- Permissions and Access Control
- Token-Based Authentication
- Obtaining and Using Tokens
- Other Token Types
- Summary

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ModelSerializers in DRF

A `ModelSerializer` in Django REST Framework (DRF) automatically generates serializer fields based on a model, similar to how a `ModelForm` works.

Suppose you have a `Book` model:

```
from rest_framework import serializers
from .models import Book

class BookSerializer(serializers.ModelSerializer):
    class Meta:
        model = Book
        fields = '__all__'
```

- This serializer will include all fields from the `Book` model.
- If you add new fields to `Book`, they are automatically included.
- You can specify fields explicitly if you want more control.

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Routers and URL Patterns

DRF routers can automatically generate URL patterns for your viewsets, reducing boilerplate code.

Registering a `BookViewSet` with a router:

```
from rest_framework.routers import DefaultRouter
from .views import BookViewSet

router = DefaultRouter()
router.register(r'books', BookViewSet)

urlpatterns = [
    # ...other patterns...
] + router.urls
```

- The router creates endpoints like `/books/` and `/books/{id}/`.
- This approach avoids manual URL pattern definitions for each CRUD operation.
- You can still add custom endpoints alongside router-generated ones.

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Permissions and Access Control

DRF permissions control who can access your API endpoints.

Allowing only authenticated users to access a viewset:

```
from rest_framework.permissions import IsAuthenticated

class BookViewSet(viewsets.ModelViewSet):
    permission_classes = [IsAuthenticated]
    queryset = Book.objects.all()
    serializer_class = BookSerializer
```

- `IsAuthenticated` restricts access to users who are logged in.
- You can use other permission classes for different access rules.
- Permissions can be set globally or per-view.

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Token-Based Authentication

Token authentication allows clients to authenticate using a token instead of a username and password.

Setting up token authentication in `settings.py`:

```
INSTALLED_APPS = [
    # ... other apps ...
    'rest_framework.authtoken',
]

REST_FRAMEWORK = {
    'DEFAULT_AUTHENTICATION_CLASSES': [
        'rest_framework.authentication.TokenAuthentication',
    ],
}
```

- Tokens are generated per user and sent in the request header.
- This is useful for mobile apps or third-party clients.
- Tokens can be managed via the admin panel or programmatically.

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Obtaining and Using Tokens

Clients obtain tokens by sending credentials to a special endpoint, then use the token for future requests.

Adding a token endpoint to your URLs:

```
from rest_framework.authtoken.views import obtain_auth_token

urlpatterns = [
    # ... other patterns ...
    path('api/token-auth/', obtain_auth_token, name='api_token_auth'),
]
```

- Clients POST their username and password to `/api/token-auth/` to get a token.
- Future requests include `Authorization: Token <token>` in the header.
- Without a valid token, protected endpoints return a permission denied error.

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Other Token Types

Beyond DRF's default tokens, you can use JWT or OAuth2 for more advanced authentication.

JWT header example:

```
Authorization: Bearer <your_jwt_token>
```

- JWT and OAuth2 support features like token expiration and refresh.
- Use packages like `djangorestframework-simplejwt` or `django-oauth-toolkit` for these methods.
- Choose the token type that fits your application's security needs.

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Summary

- DRF provides powerful tools for building REST APIs with Django.
- ModelSerializers simplify data serialization.
- ModelViewSets and routers reduce the amount of code needed for CRUD operations.
- Routers reduce URL configuration overhead.
- Permissions and token authentication enhance security.
- Choose the right authentication method based on your application’s requirements.

Example

Let’s go do an example together